InsurCalim AI - Business Case

InsurCalim AI — Business Case & Investment Brief

An AI platform for investigating auto-insurance claims: explainable, human-in-the-loop, and built to attack a multi-hundred-billion-dollar fraud problem.

Prepared for review · MVP working end-to-end · synthetic & public data only (no real PII).

---

1. Executive Summary

Insurance claims are slow, expensive, and heavily exposed to fraud. Every claim arrives as a pile of unstructured documents, photos, and videos that a human adjuster must read, cross-check, and price — millions of times a year. Meanwhile, fraud drains **hundreds of billions of dollars annually** and is now being supercharged by AI-generated fake images and documents.

InsurCalim AI is a working multi-agent platform that ingests a claim's documents, photos, and videos and produces a **ranked, explainable investigator report** — risk score, event timeline, flagged inconsistencies, coverage decision, a fair-cost (quantum) assessment, citations, and a recommended action. It is **human-in-the-loop by design: it flags and ranks, it never auto-denies.**

The MVP is complete and runs end-to-end. This document sets out the problem size, the market opportunity, the solution, its architecture, its **operating cost**, and what funding unlocks.

The opportunity in one line: the AI-in-insurance market is growing at **~32–35% CAGR** toward **$88–246B by 2030–2035**, and AI fraud detection can cut fraud losses by **up to 30%** — while our cost to analyse a claim can be **as low as a few cents**.

---

2. The Problem and Its Size

Claims handling is the costliest, most fraud-exposed part of insurance, and it is still largely manual.

- **Fraud is enormous.** Global insurance fraud is estimated to exceed **80*billionperyear* * *(2025).*IntheUnitedStatesalone,frauddrainsanestimated* * *308.6 billion per year**, of which **$90–122 billion** comes from Property & Casualty (P&C) lines. [1][2]
- **It is widespread.** Fraud occurs in roughly **10% of P&C insurance losses**, and it costs the average US consumer about **$900 per year** in higher premiums. [1][2]
- **It is getting worse with AI.** An estimated **25–30% of claims** now involve GenAI-altered fake images, medical reports, or valuation certificates — a threat traditional tooling was never built to

catch. [1]
- **Manual processing is slow and expensive.** Adjusters open individual scans and photos, analyse unstructured files, and re-key data into legacy systems — a bottleneck that limits growth and delays honest payouts. [3][4]

Why this matters for InsurCalim AI: the platform is built precisely around these pain points — multi-format document intake, multi-signal fraud detection (including **AI-generated media detection**), and an explainable score an investigator can defend.

---

3. Market Opportunity

Insurance is one of the largest and fastest-moving arenas for applied AI, and every major player is investing.

- **AI in insurance** was valued at roughly **10–20*billionin*2025 * **andisprojectedtoreach* * **88–246 billion by 2030–2035**, a CAGR of **~32–35%**. [5][6]
- **AI in claims processing** specifically is forecast to grow from **$0.46B (2025) to $0.97B by 2030** (16.2% CAGR). [6]
- **AI fraud detection** could reduce insurance fraud losses by **up to 30%**, translating to billions in savings. [6]

The field is crowded with serious money and proven results, which validates the category:

Player	What they did	Result
EY (Nordic insurer)	Document Intelligence (OCR + NLP)	**70%** auto-extraction [3]
PwC (auto insurer)	Explainable-AI damage assessment	**29%** efficiency savings [7]
McKinsey / Aviva	80+ AI models, human-in-the-loop claims	**£60M** saved in 2024 [8]
Shift Technology	Fraud & underwriting detection	**$30M+/yr** risk mitigation [9]
AWS / EXL	GenAI underwriting on Bedrock	**−80%** underwriting cost [10]
Lemonade	AI-native claims ("AI Jim")	**~3-second** settlements [11]

InsurCalim AI applies the same patterns these leaders converged on — document intelligence, explainability, and human-in-the-loop control — in a single, coherent, portfolio-ready product.

---

4. Our Solution — InsurCalim AI

A multi-agent system that turns a raw claim into a defensible investigator report.

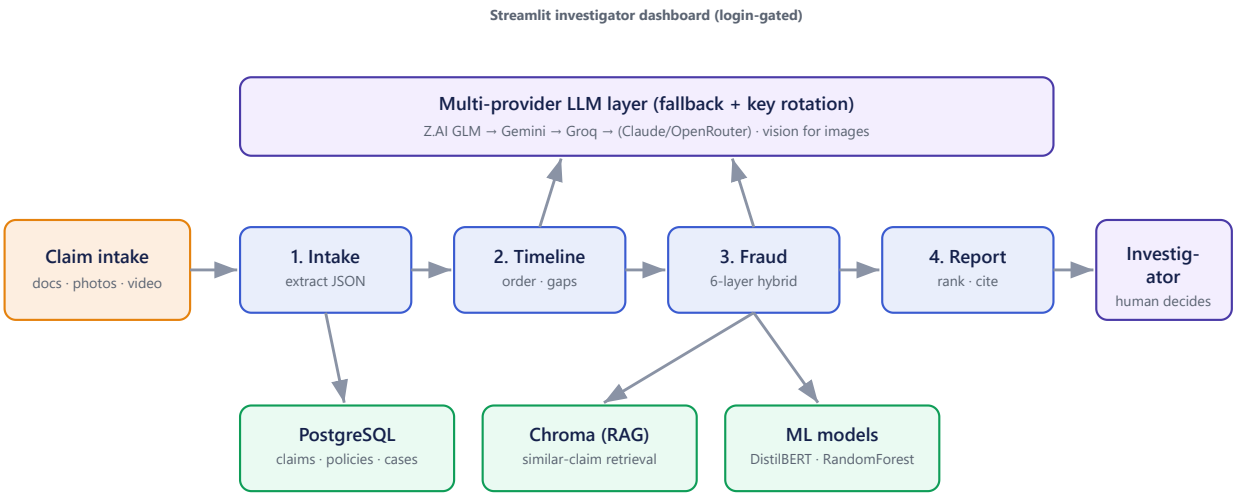
- 🧠 **Four specialised agents** orchestrated by a LangGraph state machine: `intake → timeline → fraud → report`.
- 🗂️ **Multi-format intake** — TXT, PDF, DOCX, images, and video, with a vision-OCR fallback for scanned/image-only PDFs.

- 🕵️ **Multi-signal fraud detection** — deterministic rules, duplicate-photo hashing, a fine-tuned ML score, similar-past-claim retrieval (RAG), full-document narrative red-flag analysis, and **AI-generated (deepfake) media detection**.
- 🛡️ **Policy-driven coverage gate** — checks coverage and whether every policy-required document is present.
- 💰 **Claim-amount (quantum) assessment** — even when a claim is not fraud, it judges whether the demanded cost is justified and, if not, estimates the fair amount.
- 📄 **Claimant overview** — a consolidated profile, summary, document checklist, and risk snapshot per uploaded claimant.
- 📄 **Explainable, cited report** — a risk score that breaks down into weighted reason codes, with evidence and policy-clause citations; **human-in-the-loop — it never auto-denies.**
- 💬 **AI chat** — grounded, per-claim follow-up questions for the investigator.

---

5. Architecture

A shared `ClaimState` flows through the agent graph; supporting stores and models sit alongside.



Flow: a claim enters intake, is extracted to structured JSON, ordered into a timeline, scored for fraud across six layers, then synthesised into a ranked, cited report — with a human investigator making the final call. PostgreSQL, Chroma (RAG), and the ML models support the agents; a resilient multi-provider LLM layer powers extraction and reasoning; everything is served through a login-gated Streamlit dashboard.

---

6. Technology Stack

Layer	Technology
Orchestration	LangGraph · LangChain-core

Layer	Technology
LLMs	Multi-provider fallback + key rotation: **Z.AI (GLM)** → **OpenRouter** → **Gemini** → **Claude** → **Groq**; Gemini/OpenRouter for vision
Vector DB / RAG	Chroma (Pinecone-ready abstraction)
Relational store	PostgreSQL (claims, parties, vehicles, photo hashes, policies, cases)
ML / fine-tuning	DistilBERT (narrative) · class-balanced Random Forest (tabular, real 30k rows)
Frontend	Streamlit dashboard, custom CSS, login-gated
Media	pypdf · pdfplumber · pypdfium2 (vision-OCR) · python-docx · OpenCV · Pillow · imagehash
Infra	Docker Compose · AWS deployment plan (RDS · S3 · ECS/EC2, optional SageMaker / Bedrock)

7. Cost Analysis — What It Costs to Analyse a Claim

This is the core operating-cost question: *how much does it cost to run document/claim analysis?*

Per-claim workload. A single claim runs roughly **7–8 text LLM calls** (intake extraction, timeline, two fraud passes, report synthesis, coverage, amount assessment, document completeness) plus **one vision call per image**. That is approximately **20,000–30,000 input tokens** and **~5,000 output tokens** per claim in total.

Cost per claim by provider (using ~25K input + 5K output tokens):

Provider (per 1M tokens in/out)	Cost per claim	Per 1,000 claims
Z.AI GLM-4.5-flash / Groq — free tier *(current default)*	**~$0.00**	**~$0**
Gemini 2.5 Flash ($0.30 / $2.50) [12]	**~$0.02**	**~$20**
Claude Sonnet 5 ($3 / $15) [13]	**~$0.15**	**~$150**
Claude Opus 4.8 ($5 / $25) [13]	**~$0.25**	**~$250**

> Illustrative math (Gemini Flash): 25K × $0.30/1M = $0.0075 input + 5K × $2.50/1M = 0.0125*output* ≈ **0.02/claim**. Vision-heavy claims add a small per-image increment.

Run-rate scenarios (add fixed infrastructure — managed Postgres + app hosting ≈ **$50–150 / month** at pilot scale):

Scenario	LLM tier	10,000 claims/month	+ infra	Monthly total
Lean / demo	free tier	~$0	~$50	**~$50**

Scenario	LLM tier	10,000 claims/month	+ infra	Monthly total
Balanced	Gemini Flash	~$200	~$100	**~$300**
Premium	Claude Sonnet	~$1,500	~$150	**~$1,650**

The ROI framing. Fraud costs the average consumer **~$900/year** and AI fraud detection can cut fraud losses by **up to 30%**. [1][6] Even at the premium tier, analysing a claim costs **cents**, while catching a single inflated or fraudulent claim saves **thousands** — an order-of-magnitude return. The platform is deliberately built to start at near-zero cost on free tiers and scale up to paid, higher-accuracy models only where the ROI justifies it.

7.1 Free vs. paid — the quality trade-off

Free-tier models produce acceptable extraction and summaries, but the **hardest and most valuable task — reading the whole claim file and catching where different documents contradict each other** (dates, damage descriptions, amounts) — is a deep-reasoning task where small free models fall short. This is exactly where premium models earn their cost:

Tier	What it catches	Typical use
Free (GLM / Llama / Flash-free)	Extraction, summaries, obvious red flags. Misses subtle cross-document contradictions.	Demo, low-stakes triage
Claude Sonnet 5	Reasons over the full document set; reliably flags "document A says X, document B says Y"; much stronger summaries. **The sweet spot.**	Production
Claude Opus 4.8	Sonnet's quality plus the hardest, most ambiguous cases; deepest reasoning (incremental gain over Sonnet on most claims).	High-stakes / escalated

The smart architecture is a hybrid, not one model for everything: keep cheap models for the mechanical steps (intake, timeline, coverage, amount, documents) and spend on a premium model **only for the reasoning-heavy fraud + report step** — where contradiction detection lives.

Setup	Cost / claim	Per 1,000	~10,000 / month
Free tier (today)	~$0	~$0	~$0
Hybrid — Flash + Sonnet for reasoning *(recommended)*	~$0.07	~$70	~$700
Hybrid — Flash + Opus for reasoning	~$0.12	~$120	~$1,200
All Sonnet / all Opus	$0.15 / $0.25	$150 / $250	$1,500 / $2,500

The hybrid delivers roughly **90% of premium quality at under half the cost** — the recommended path to production.

On training. For contradiction detection and summarisation, a strong base model with good prompting **beats fine-tuning a small model**, and avoids the data-collection, labelling, and GPU cost — so early funding is best spent on a **better base model, not on training an LLM**. Fine-tuning is reserved for the

numeric fraud classifier (the existing DistilBERT / Random Forest layer), where more labelled real-world claims data directly improves the score — a Phase-2 investment once a pilot supplies that data.

---

8. Roadmap & The Ask

The MVP is complete and working end-to-end. Funding accelerates the path from demo to pilot:

- **Paid LLM tier** — move the primary path to a paid, higher-accuracy model for production reliability and consistency (a modest, usage-based cost per the table above).
- **Cloud deployment** — AWS (RDS · S3 · ECS), so an insurer can pilot on real infrastructure.
- **Pilot with real (permissioned) data** — validate accuracy and fraud-catch rate on a partner insurer's historical claims.
- **Hardening** — output-consistency controls, audit logging, and role-based access for multi-user investigator teams.

The ask: seed funding to run a paid-tier pilot and a cloud deployment, turning a working MVP into a revenue-ready product in a market growing at ~32–35% per year.

---

9. References

1. Coalition Against Insurance Fraud / industry statistics — *Insurance fraud worldwide and US costs, P&C share, consumer impact, GenAI fake-media trend.* https://insurancefraud.org/fraud-stats/ ; https://www.iii.org/fact-statistic/facts-and-statistics-insurance-fraud
2. Insurance Information Institute — *Facts + Statistics: Fraud.* https://www.iii.org/fact-statistic/facts-and-statistics-insurance-fraud
3. EY — *How a Nordic insurance company automated claims processing.* https://www.ey.com/en_dk/insights/financial-services/emeia/how-a-nordic-insurance-company-automated-claims-processing
4. InsurCalim AI — companion research survey, `docs/research/ai-in-insurance-claims.md`.
5. Fortune Business Insights — *AI in Insurance Market Size, Share.* https://www.fortunebusinessinsights.com/ai-in-insurance-market-114760
6. *AI in Insurance Claims Processing Market* (GlobeNewswire) and Mordor Intelligence — market size, CAGR, and fraud-loss reduction. https://www.globenewswire.com/news-release/2026/07/06/3322187/28124/en/AI-in-Insurance-Claims-Processing-Market-Surges-to-0-97-Billion-by-2030-with-a-CAGR-of-16-2.html ; https://www.mordorintelligence.com/industry-reports/ai-in-insurance-market
7. PwC — *Auto insurance AI analytics case study (explainable AI damage estimation).* https://www.pwc.com/us/en/library/case-studies/auto-insurance-ai-analytics.html
8. McKinsey & Company — *Aviva: Rewiring the insurance claims journey with AI.* https://www.mckinsey.com/capabilities/tech-and-ai/how-we-help-clients/rewired-in-action/aviva-rewiring-the-insurance-claims-journey-with-ai
9. Shift Technology — *AI in underwriting: case examples from leading insurers.* https://www.shift-technology.com/resources/reports-and-insights/ai-in-underwriting-case-examples-from-leading-insurers

10. AWS / EXL — *EXL reduces underwriting costs by up to 80% using Amazon Bedrock.* https://aws.amazon.com/solutions/case-studies/exl-case-study/
11. Lemonade — *AI Jim claims automation.* https://www.claimsjournal.com/news/national/2025/03/07/327670.htm
12. Google — *Gemini 2.5 Flash API pricing ($0.30 input / $2.50 output per 1M tokens).* https://ai.google.dev/pricing
13. Anthropic — *Claude API pricing (Sonnet 5 3/15; Opus 4.8 5/25 per 1M tokens).* https://platform.claude.com/docs/en/pricing

---

InsurCalim AI is a portfolio project built on synthetic and public data only — no real customer information. All market and pricing figures are attributed to the public sources cited above. Decision support for a human investigator, not an automated decision.